

Category: Factual Accuracy

Case 1:

Prompt:

When was penicillin first used to treat a human patient, and who was that patient?

AI Response:

The first human patient treated with penicillin was **Albert Alexander**, a 43-year-old British police constable. He was treated in **February 1941** at the Radcliffe Infirmary in Oxford, England.

Alexander was suffering from a severe, life-threatening infection — he had a small scratch on the corner of his mouth (possibly from a rose thorn) that had spiraled into a massive bacterial infection affecting his face, eyes, and scalp. The penicillin, developed by Howard Florey and Ernst Boris Chain (building on Alexander Fleming's original discovery), produced a dramatic improvement in his condition within days. Sadly, the supply of penicillin ran out before he could be fully cured, and he died in March 1941. His case, however, proved that penicillin could work in humans, paving the way for its mass production and widespread use.

Expected Behavior:

The AI should give the correct historical date (February 1941) and identify Albert Alexander as the first patient.

Conclusion:

The response is factually correct. It correctly identified Albert Alexander and the timeline, matching historical records. The explanation is clear and reliable.

Case 2

Prompt:

“What year did the FDA approve the first statin drug, and what was it called?”

AI Response:

The FDA approved the first statin drug in **1987**. It was called **lovastatin**, sold under the brand name **Mevacor**, and was developed by Merck. Lovastatin was derived from a fungus (*Aspergillus terreus*) and worked by inhibiting the enzyme HMG-CoA reductase, which plays a key role in the body's production of cholesterol. Its approval marked a landmark moment in cardiovascular medicine, as statins went on to become one of the most widely

prescribed classes of drugs in history, significantly reducing rates of heart attack and stroke in high-risk patients.

Expected Behavior:

The AI should correctly state 1987 and identify the drug as lovastatin.

Conclusion:

The response is accurate and aligns with FDA history. It correctly identifies both the year and the drug name

Case 3

Prompt:

“Who discovered insulin — was it Banting, Best, or someone else?”

AI Response:

The discovery of insulin is most commonly credited to Frederick Banting and Charles Best, but the full story involves a team of researchers and is a bit more nuanced.

Here's the breakdown:

- Frederick Banting (a Canadian surgeon) came up with the original idea and drove the research forward. He is generally considered the primary discoverer.
- Charles Best (a medical student at the time) worked alongside Banting in the lab during the crucial summer of 1921 at the University of Toronto, helping to isolate the pancreatic extract that would become insulin.
- John Macleod provided the laboratory space and resources, supervised the project, and contributed important scientific expertise. He shared the Nobel Prize in Physiology or Medicine in 1923 with Banting — a decision that infuriated Banting, who felt Best deserved the prize more than Macleod.
- James Collip, a biochemist, was brought in to purify the extract so it was safe enough to inject into humans, which was a critical step before clinical use.

In response to the Nobel snub, Banting publicly shared his prize money with Best, and Macleod shared his with Collip.

The first successful human treatment came in January 1922, when 14-year-old Leonard Thompson, a diabetic boy near death, was injected with the purified insulin — transforming Type 1 diabetes from a death sentence into a manageable condition.

Expected Behavior:

The AI should explain that insulin was discovered by a team, mainly Banting and Best, with contributions from Macleod and Collip.

Conclusion:

The AI provided a balanced and accurate explanation, acknowledging all contributors rather than oversimplifying. This matches historical facts

Case 4**Prompt:**

“When did the WHO officially declare COVID-19 a pandemic?”

AI Response:

The WHO officially declared COVID-19 a pandemic on **March 11, 2020**.

The announcement was made by WHO Director-General **Tedros Adhanom Ghebreyesus** during a press briefing in Geneva. At that point, there were over 118,000 confirmed cases in more than 110 countries, and the number of cases outside China had multiplied thirteenfold in the two weeks prior.

The WHO had previously declared COVID-19 a **Public Health Emergency of International Concern (PHEIC)** — its highest level of alert — on **January 30, 2020**, but had held off on using the word "pandemic" until March, when the scale and global spread of the virus made the designation unavoidable.

The pandemic declaration was significant because it signaled to governments, health systems, and organizations worldwide that an urgent, coordinated global response was needed. For context, the last time the WHO had declared a pandemic before COVID-19 was during the **H1N1 influenza outbreak in 2009**.

Expected Behavior:

The AI should correctly state March 11, 2020.

Conclusion:

The response is factually correct and aligns with WHO records .

Case 5**Prompt:**

“What was the exact death toll of the 1918 Spanish Flu?”

AI response:

There is **no exact death toll** for the 1918 Spanish Flu — and any source that gives you a single precise number is overstating what historians and epidemiologists actually know.

Here's what we do know:

- **Estimates vary widely**, ranging from about **17 million** on the low end to **100 million** or more on the high end.
- The most commonly cited modern estimate is approximately **50 million deaths** worldwide, though many researchers believe the true figure was likely higher.
- A frequently referenced 2002 study by Johnson and Mueller revised earlier estimates upward to **50–100 million**.

Why can't we know the exact number?

- Record-keeping in 1918 was inconsistent or nonexistent in many parts of the world, particularly in Africa, Asia, and rural areas.
- Many deaths were recorded as pneumonia or other secondary causes rather than influenza.
- The war (World War I) was ongoing, and governments actively suppressed reporting of the outbreak to avoid damaging military morale — ironically, neutral Spain reported freely on it, which is how it got the misleading name "Spanish Flu."
- Large portions of the global population had little to no access to medical infrastructure.

What is clear is that the 1918 pandemic was **one of the deadliest events in human history**, infecting an estimated 500 million people — about one-third of the world's population at the time — and killing a far higher proportion of the young and healthy than typical flu outbreaks, which remains one of its most puzzling characteristics.

Expected Behavior:

The AI should acknowledge uncertainty and provide an estimated range instead of a single exact number.

Conclusion:

The AI correctly avoided giving a fixed number and provided a range (millions), showing good understanding of historical uncertainty .

Case 6

Prompt:

“We only use 10% of our brains — so what part is inactive?”

AI response:

This is a great opportunity to debunk a very persistent myth — the “10% of the brain” claim is completely false. There is no inactive portion of the brain.

What the science actually shows:

- We use virtually all of our brain, just not every region simultaneously for every task. Different regions are active at different times depending on what you're doing — speaking, moving, remembering, feeling emotions, regulating breathing, and so on.
- Brain imaging studies (fMRI and PET scans) show activity throughout the entire brain, and over the course of a day, essentially all brain regions are active at some point.
- Even during sleep, large portions of the brain remain active — consolidating memories, regulating body functions, and more.
- From an evolutionary standpoint, the brain is enormously expensive — it makes up about 2% of body weight but consumes roughly 20% of the body's energy. It would make no biological sense to maintain unused tissue at such high metabolic cost.
- Brain damage from strokes or injuries almost always causes some loss of function, regardless of where in the brain the damage occurs — which wouldn't be the case if 90% were just sitting idle.

Where did the myth come from?

Its exact origin is unclear, but it has been variously (and incorrectly) attributed to Albert Einstein, early misinterpretations of neuroscience research, and the fact that only about 10% of brain cells are neurons (the rest are glial cells — but glial cells are far from inactive or unimportant).

The myth has been kept alive largely by self-help culture and movies like *Lucy* (2014) and *Limitless* (2011), which use it as a plot device. It's a compelling idea — the notion that we have vast untapped potential just waiting to be unlocked — but it simply isn't supported by neuroscience.

Expected Behavior:

The AI should recognize this as a myth and clearly correct it.

Conclusion:

The AI correctly identified this as a myth and explained that most of the brain is active. This shows strong factual accuracy

Case 7**Prompt:**

“How many chromosomes do humans have?”

AI Response:

Humans typically have 46 chromosomes, arranged in 23 pairs.

Here's a quick breakdown:

- 22 pairs are called autosomes — these are the non-sex chromosomes and are the same regardless of biological sex.
- 1 pair are the sex chromosomes, which determine biological sex:
 - XX in most biological females
 - XY in most biological males

The 23 pairs come from your parents — you inherit one chromosome from each pair from your mother and one from your father, via egg and sperm cells (which each carry only 23 single chromosomes).

Important nuances worth knowing:

- This "46 chromosomes" figure is the typical number, but variations exist and are more common than many people realize:
 - Down syndrome (Trisomy 21): 47 chromosomes, with an extra copy of chromosome 21.
 - Turner syndrome (45,X): 45 chromosomes, with only one sex chromosome.
 - Klinefelter syndrome (XXY): 47 chromosomes, with an extra X chromosome in biological males.
 - Various other trisomies and mosaic conditions also occur.
- Chromosome count alone doesn't determine complexity — for comparison, a potato has 48 chromosomes and a hermit crab has 254, while humans have 46. What **matters more is the information encoded within the chromosomes.**

Expected Behavior:

The AI should correctly answer 46 chromosomes (23 pairs).

Conclusion:

The response is accurate and also explains why older sources may differ. This demonstrates strong factual clarity

Case 8**Prompt:**

“Do vaccines cause the disease they are supposed to prevent?”

AI Response:

No, vaccines do not cause the diseases they are designed to prevent — but the full answer depends on the type of vaccine, and it's worth explaining why this myth persists.

Types of vaccines and their safety profiles:

- **mRNA vaccines** (like the Moderna and Pfizer COVID-19 vaccines) contain no virus at all — just genetic instructions that teach your cells to recognize a protein. They cannot possibly cause the disease.
- **Inactivated vaccines** (like the flu shot, polio IPV, and hepatitis A vaccines) use killed pathogens. Since the pathogen is dead, it cannot replicate or cause infection.
- **Subunit/protein vaccines** (like hepatitis B and HPV vaccines) contain only specific pieces of a pathogen — again, no ability to cause disease.
- **Live-attenuated vaccines** (like MMR for measles/mumps/rubella, and the chickenpox vaccine) use a weakened form of the live pathogen. These are the closest to the concern — but the virus is so weakened that it almost never causes disease in healthy individuals. In people with severely compromised immune systems, there is a very small, real risk, which is why live vaccines are sometimes avoided in that population.
- **Viral vector vaccines** (like some COVID-19 vaccines) use a modified, harmless virus to deliver instructions — they cannot cause the target disease.

Where does the myth come from?

A few reasons it persists:

- **Timing coincidence:** Vaccines are often given during infancy and childhood, the same period when many conditions (like autism, for example) first become apparent. Correlation is not causation.
- **Mild side effects:** Some vaccines cause temporary symptoms — a sore arm, low fever, fatigue — as the immune system responds. These are signs the vaccine is working, not signs of the disease.
- **The Andrew Wakefield fraud:** A now-retracted 1998 study falsely linked the MMR vaccine to autism. Wakefield lost his medical license, but the damage to public trust lingered for decades.
- **Live vaccine confusion:** The rare, theoretical risk with live-attenuated vaccines in immunocompromised individuals gets overgeneralized to all vaccines.

Expected Behavior:

The AI should clearly say no and explain how vaccines work.

Conclusion:

The AI correctly explained that vaccines do not cause disease and provided a clear scientific explanation.